composed of some of the molecules implicated in the origin of life on Earth four billion years ago.

In local winter, layers of ice and snow build up on the surface. (Our winters, mercifully, are only 4 percent as long.) Through the spring, they are slowly transformed, more and more reddish organic molecules accumulating. By summertime, the ice and snow have evaporated; the gases so released migrate halfway across the planet to the winter hemisphere and there cover the surface with ice and snow again. But the reddish organic molecules do not vaporize and are not transported—a lag deposit, they are next winter covered over by new snows, which are in turn irradiated, and by the following summer the accumulation is thicker. As time goes on, substantial amounts of organic matter are built up on the surface of Triton, which may account for its delicate color markings.

The streaks begin in small, dark source regions, perhaps when the warmth of spring and summer heats subsurface volatile snows. As they vaporize, gas comes gushing out as in a geyser, blowing off less-volatile surface snows and dark organics. Prevailing low-speed winds carry away the dark organics, which slowly sediment out of the thin air, are deposited on the ground, and generate the appearance of the streaks. This, at least, is one reconstruction of recent Tritonian history.

Triton may have large, seasonal polar caps of smooth nitrogen ice underlying layers of dark organic materials. Nitrogen snows seem recently to have fallen at the equator. Snowfalls, geysers, windblown organic dust, and high-altitude hazes were entirely unexpected on a world with so thin an atmosphere.

Why is the air so thin? Because Triton is so far from the Sun. Were you somehow to pick this world up and move it into orbit around Saturn, the nitrogen and methane ices would quickly evaporate, a much denser atmosphere of gaseous nitrogen and methane would form, and radiation would generate an opaque tholin haze. It would become a world very like Titan. Conversely, if you moved Titan into orbit about Neptune, almost all its atmosphere would freeze out as snows and ices, the tholin would fall out and